# Asterix category 062 - SDPS Track Messages

**category**: 062

**edition**: 1.20

**date**: 2023-02-13

## Preamble

Surveillance data exchange.

## Description of standard data items

### I062/010 - Data Source Identifier

definition: Identification of the system sending the data.

Group

#### I062/010/SAC - System Area Code

Element
bit size: 8
Raw Content

#### I062/010/SIC - System Identification Code

Element
bit size: 8
Raw Content

**Note:**

- The up-to-date list of SACs is published on the EUROCONTROL Web Site (http://www.eurocontrol.int/asterix).

### I062/015 - Service Identification

definition: Identification of the service provided to one or more users.

Element
bit size: 8
Raw Content

### I062/040 - Track Number

definition: Identification of a track.

Element
bit size: 16
Raw Content

### I062/060 - Track Mode 3/A Code

definition: Mode-3/A code converted into octal representation.

Group

#### I062/060/V - Validated

Element
bit size: 1
Values:

**0:** Code validated
**1:** Code not validated

**I062/060/G - Garbled**

Element
bit size: 1
Values:

**0:** Default
**1:** Garbled code

**I062/060/CH - Change in Mode 3/A**

Element
bit size: 1
Values:

**0:** No change
**1:** Mode 3/A has changed

Spare bits: 1

**I062/060/MODE3A - Mode-3/A Reply in Octal Representation**

Element
bit size: 12
Octal string (3-bits per char)

### I062/070 - Time Of Track Information

definition: Absolute time stamping of the information provided in the track message, in the form of elapsed time since last mid night, expressed as UTC.

Element
bit size: 24
Unsigned quantity
LSB = $1/2^7$ s $\approx 7.8125e-3$ s
unit: ”s”

Notes:

1. This is the time of the track state vector.
2. The time is reset to zero at every midnight.

### I062/080 - Track Status

definition: Status of a track.

Extended

**I062/080/MON**

Element
bit size: 1
Values:

**0:** Multisensor track
**1:** Monosensor track

**I062/080/SPI**

Element
bit size: 1
Values:

**0:** Default value

**1:** SPI present in the last report received from a sensor capable of decoding this data

**I062/080/MRH - Most Reliable Height**

Element
bit size: 1
Values:

**0:** Barometric altitude (Mode C) more reliable
**1:** Geometric altitude more reliable

**I062/080/SRC - Source of Calculated Track Altitude for I062/130**

Element
bit size: 3
Values:

**0:** No source
**1:** GNSS
**2:** 3D radar
**3:** Triangulation
**4:** Height from coverage
**5:** Speed look-up table
**6:** Default height
**7:** Multilateration

**I062/080/CNF**

Element
bit size: 1
Values:

**0:** Confirmed track
**1:** Tentative track

*(FX) - extension bit*

**I062/080/SIM**

Element
bit size: 1
Values:

**0:** Actual track
**1:** Simulated track

**I062/080/TSE**

Element
bit size: 1
Values:

**0:** Default value
**1:** Last message transmitted to the user for the track

**I062/080/TSB**

Element
bit size: 1
Values:

**0:** Default value
**1:** First message transmitted to the user for the track

**I062/080/FPC**

Element
bit size: 1
Values:

**0:** Not flight-plan correlated
**1:** Flight plan correlated

**I062/080/AFF**

Element
bit size: 1
Values:

- **0:** Default value
- **1:** ADS-B data inconsistent with other surveillance information

**I062/080/STP**

Element
bit size: 1
Values:

- **0:** Default value
- **1:** Slave Track Promotion

**I062/080/KOS**

Element
bit size: 1
Values:

- **0:** Complementary service used
- **1:** Background service used

*(FX) - extension bit*

**I062/080/AMA**

Element
bit size: 1
Values:

- **0:** Track not resulting from amalgamation process
- **1:** Track resulting from amalgamation process

**I062/080/MD4**

Element
bit size: 2
Values:

- **0:** No Mode 4 interrogation
- **1:** Friendly target
- **2:** Unknown target
- **3:** No reply

**I062/080/ME**

Element
bit size: 1
Values:

- **0:** Default value
- **1:** Military Emergency present in the last report received from a sensor capable of decoding this data

**I062/080/MI**

Element
bit size: 1
Values:

- **0:** Default value
- **1:** Military Identification present in the last report received from a sensor capable of decoding this data

**I062/080/MD5**

Element
bit size: 2
Values:

- **0:** No Mode 5 interrogation
- **1:** Friendly target

**2:** Unknown target
**3:** No reply

*(FX) - extension bit*

**I062/080/CST**

Element
bit size: 1
Values:
**0:** Default value
**1:** Age of the last received track update is higher than system dependent threshold (coasting)

**I062/080/PSR**

Element
bit size: 1
Values:
**0:** Default value
**1:** Age of the last received PSR track update is higher than system dependent threshold

**I062/080/SSR**

Element
bit size: 1
Values:
**0:** Default value
**1:** Age of the last received SSR track update is higher than system dependent threshold

**I062/080/MDS**

Element
bit size: 1
Values:
**0:** Default value
**1:** Age of the last received Mode S track update is higher than system dependent threshold

**I062/080/ADS**

Element
bit size: 1
Values:
**0:** Default value
**1:** Age of the last received ADS-B track update is higher than system dependent threshold

**I062/080/SUC**

Element
bit size: 1
Values:
**0:** Default value
**1:** Special Used Code (Mode A codes to be defined in the system to mark a track with special interest)

**I062/080/AAC**

Element
bit size: 1
Values:
**0:** Default value
**1:** Assigned Mode A Code Conflict (same discrete Mode A Code assigned to another track)

*(FX) - extension bit*

**I062/080/SDS**

Element
bit size: 2
Values:

- **0:** Combined
- **1:** Co-operative only
- **2:** Non-Cooperative only
- **3:** Not defined

**I062/080/EMS**

Element
bit size: 3
Values:

- **0:** No emergency
- **1:** General emergency
- **2:** Lifeguard / medical
- **3:** Minimum fuel
- **4:** No communications
- **5:** Unlawful interference
- **6:** Downed Aircraft
- **7:** Undefined

**I062/080/PFT**

Element
bit size: 1
Values:

- **0:** No indication
- **1:** Potential False Track Indication

**I062/080/FPLT**

Element
bit size: 1
Values:

- **0:** Default value
- **1:** Track created / updated with FPL data

*(FX) - extension bit*

**I062/080/DUPT**

Element
bit size: 1
Values:

- **0:** Default value
- **1:** Duplicate Mode 3/A Code

**I062/080/DUPF**

Element
bit size: 1
Values:

- **0:** Default value
- **1:** Duplicate Flight Plan

**I062/080/DUPM**

Element
bit size: 1
Values:

- **0:** Default value
- **1:** Duplicate Flight Plan due to manual correlation

**I062/080/SFC**

Element
bit size: 1
Values:

**0:** Default value
**1:** Surface target

**I062/080/IDD**

Element
bit size: 1
Values:

**0:** No indication
**1:** Duplicate Flight-ID

**I062/080/IEC**

Element
bit size: 1
Values:

**0:** Default value
**1:** Inconsistent Emergency Code

**I062/080/MLAT**

Element
bit size: 1
Values:

**0:** Default value
**1:** Age of the last received MLAT track updateis higher than system dependent threshold

*(FX) - extension bit*

Notes:

1. Track type and coasting can also be derived from I062/290 System Track Update Ages
2. If the system supports the technology, default value (0) means that the technology was used to produce the report
3. If the system does not support the technology, default value is meaningless.
4. Bits 6/4 (EMS): other than subfield #11 of data item I062/380, these bits allow the SDPS to set the emergency indication as derived from other sources than ADS-B (e.g. based on the Mode 3/A code).
5. Bits 6/4 (EMS): if EMS is populated from ADS-B information the following shall apply: In ADS-B Version 3 (as defined in I021/210/VN) some values of EMS have been redefined. In order to provide the information also in Data Item I062/080/EMS, mapping is required to ensure that information is not lost in systems not yet capable to decode this Edition of Category 062. If I021/210/VN = 3, the values contained in I062/REF/PS3 shall be mapped to I062/380/EMS in line with the following table: :

| ADS-B Version 3 (I062/REF/PS3) | ADS-B Version < 3 (I062/080/EMS) |
|---|---|
| 0 (No Emergency/not reported) | 0 (No emergency/not reported) |
| 1 (General emergency) | 1 (General emergency) |
| 2 (UAS/RPAS Lost Link) | 4 (No communication) |
| 3 (Minimum fuel) | 3 (Minimum fuel) |
| 4 (No communication) | 4 (No communication) |
| 5 (Unlawful interference) | 5 (Unlawful interference) |
| 6 (Aircraft in distress - automatic activation) | 1 (General emergency) |
| 7 (Aircraft in distress - manual activation) | 1 (General emergency) |

6. Bit 3 (PFT): with this flag an SDPS can indicate that internal processing points to the track being potentially false. Details on the internal processing are system dependent. In order to improve security on targets provided by ADS-B numerous validation functions have been developed in the ADS-B ground domain. If any of these validation functions show a

potentially spoofed target, the PFT bit will be used to convey this information to the CWP. If and how this information is processed and displayed on the CWP is a local matter and not subject to the category 062 specification.

7. Bit 2 (FPLT): this bit - if set - indicates that the information contained in the target report has been updated by flight plan related data because no surveillance data was available for the target, or was created based on flight plan related data in areas with no surveillance.
8. Bit 8 (DUPT) is set to 1 if the correlation between the target report and a flight plan is not possible because the Mode 3/A code stated in the flight plan exists more than once in the surveillance data.
9. Bit 7 (DUPF) - if set to 1 - indicates that for a specific surveillance target more than one flight plan exists which makes correlation impossible.
10. **Bit 6 (DUPM) is set to 1 if a target was correlated manually but also a regular** flight plan exists.
11. All tracks for which bits 8, 7 or 6 are set to 1 are marked on the CWP.
12. Bit 5 (SFC) is set to 1 when the SDPS considers the target to be on the Surface (the actual meaning is implementation dependent – please refer to chapter 4.8 above).
13. Bit 4 (IDD) is set to 1 when the Flight ID is present more than once in the surveillance area.
14. Bit 3 (IEC) is set to 1 when the comparison between various sources has revealed an inconsistency in the information contained about emergency codes.
15. If I062/080 (MRH) indicates ”Barometric altitude (Mode C) more reliable”, and a calculated altitude is transmitted, it shall be transmitted using data item I062/135 “Calculated Track Barometric Altitude”.
16. If I062/080 (MRH) indicates ”Geometric altitude more reliable”, and a calculated altitude is transmitted, it shall be transmitted using data item I062/130 “Calculated Track Geometric Altitude”. In this case the source for I062/130 is indicated by I062/080 (SRC).
17. Data Items I062/130, I062/135, and I062/136 may be transmitted in parallel whenever the respective information is available. This is independent from the value transmitted on I062/080 (MRH).

**I062/100 - Calculated Track Position (Cartesian)**

definition: Calculated position in Cartesian co-ordinates with a resolution of 0.5m, in two's complement form.

Group

**I062/100/X - X Coordinate**

Element
bit size: 24
Signed quantity
LSB = $1/2$ m $\approx 0.5$ m
unit: ”m”

**I062/100/Y - Y Coordinate**

Element
bit size: 24
Signed quantity
LSB = $1/2$ m $\approx 0.5$ m
unit: ”m”

**I062/105 - Calculated Position In WGS-84 Co-ordinates**

definition: Calculated Position in WGS-84 Co-ordinates with a resolution of :math:‘180/2^{25}‘ degrees.

Group

**I062/105/LAT - Latitude**

Element
bit size: 32
Signed quantity
LSB = $180/2^{2}5$ ° ≈ $5.36441802978515625e-6$ °
unit: "°"
>= $-90.0$
<= $90.0$

**I062/105/LON - Longitude**

Element
bit size: 32
Signed quantity
LSB = $180/2^{2}5$ ° ≈ $5.36441802978515625e-6$ °
unit: "°"
>= $-180.0$
< $180.0$

Notes:

- The LSB provides a resolution at least better than 0.6m.

**I062/110 - Mode 5 Data Reports and Extended Mode 1 Code**

definition: Mode 5 Data reports and Extended Mode 1 Code.
Compound

**I062/110/SUM - Mode 5 Summary**

Group

**I062/110/SUM/M5**

Element
bit size: 1
Values:

**0:** No Mode 5 interrogation
**1:** Mode 5 interrogation

**I062/110/SUM/ID**

Element
bit size: 1
Values:

**0:** No authenticated Mode 5 ID reply
**1:** Authenticated Mode 5 ID reply

**I062/110/SUM/DA**

Element
bit size: 1
Values:

**0:** No authenticated Mode 5 Data reply or Report
**1:** Authenticated Mode 5 Data reply or Report (i.e any valid Mode 5 reply type other than ID)

**I062/110/SUM/M1**

Element
bit size: 1
Values:

**0:** Mode 1 code not present or not from Mode 5 reply
**1:** Mode 1 code from Mode 5 reply

**I062/110/SUM/M2**

Element
bit size: 1
Values:

**0:** Mode 2 code not present or not from Mode 5 reply
**1:** Mode 2 code from Mode 5 reply

**I062/110/SUM/M3**

Element
bit size: 1
Values:

**0:** Mode 3 code not present or not from Mode 5 reply
**1:** Mode 3 code from Mode 5 reply

**I062/110/SUM/MC**

Element
bit size: 1
Values:

**0:** Mode C altitude code not present or not from Mode 5 reply
**1:** Mode C altitude from Mode 5 reply

**I062/110/SUM/X - X-pulse from Mode 5 Data Reply or Report**

Element
bit size: 1
Values:

**0:** X-pulse set to zero or no authenticated Data reply or Report received
**1:** X-pulse set to one

**I062/110/PMN - Mode 5 PIN/ National Origin/Mission Code**

Group

Spare bits: 2

**I062/110/PMN/PIN - PIN Code**

Element
bit size: 14
Raw Content

Spare bits: 3

**I062/110/PMN/NAT - National Origin**

Element
bit size: 5
Raw Content

Spare bits: 2

**I062/110/PMN/MIS - Mission Code**

Element
bit size: 6
Raw Content

**I062/110/POS - Mode 5 Reported Position**

Group

**I062/110/POS/LAT - Latitude**

Element
bit size: 24
Signed quantity
LSB = $180/2^{2}3\ ° \approx 2.1457672119140625e-5\ °$
unit: ”°”
>= $-90.0$
<= $90.0$

**I062/110/POS/LON - Longitude**

Element
bit size: 24
Signed quantity
LSB = $180/2^{2}3$ ° ≈ $2.1457672119140625e-5$ °
unit: ”°”
>= $-180.0$
< $180.0$

**I062/110/GA - Mode 5 GNSS-derived Altitude**

Group

Spare bits: 1

**I062/110/GA/RES - Resolution with which the GNSS-derived Altitude (GA) is Reported**

Element
bit size: 1
Values:

**0:** GA reported in 100 ft increments
**1:** GA reported in 25 ft increments

**I062/110/GA/GA - GNSS-derived Altitude of Target, Expressed as Height Above WGS 84 Ellipsoid**

Element
bit size: 14
Signed quantity
LSB = $25$ ft ≈ $25.0$ ft
unit: ”ft”
>= $-1000.0$

**I062/110/EM1 - Extended Mode 1 Code in Octal Representation**

Group

Spare bits: 4

**I062/110/EM1/EM1 - Extended Mode 1 Reply in Octal Representation**

Element
bit size: 12
Octal string (3-bits per char)

**I062/110/TOS - Time Offset for POS and GA**

description: Time Offset coded as a twos complement number with an LSB of 1/128 s. The time at which the Mode 5 Reported Position (Subfield #3) and Mode 5 GNSS-derived Altitude (Subfield #4) are valid is given by Time of Day (I048/140) plus Time Offset.

Element
bit size: 8
Signed quantity
LSB = $1/2^{7}$ s ≈ $7.8125e-3$ s
unit: ”s”

**I062/110/XP - X Pulse Presence**

Group

Spare bits: 3

**I062/110/XP/X5 - X-pulse from Mode 5 Data Reply or Report**

Element
bit size: 1
Values:

**0:** X-pulse set to zero or no authenticated Data reply or Report received
**1:** X-pulse set to one (present)

**I062/110/XP/XC - X-pulse from Mode C Reply**

Element
bit size: 1
Values:

**0:** X-pulse set to zero or no Mode C reply
**1:** X-pulse set to one (present)

**I062/110/XP/X3 - X-pulse from Mode 3/A Reply**

Element
bit size: 1
Values:

**0:** X-pulse set to zero or no Mode 3/A reply
**1:** X-pulse set to one (present)

**I062/110/XP/X2 - X-pulse from Mode 2 Reply**

Element
bit size: 1
Values:

**0:** X-pulse set to zero or no Mode 2 reply
**1:** X-pulse set to one (present)

**I062/110/XP/X1 - X-pulse from Mode 1 Reply**

Element
bit size: 1
Values:

**0:** X-pulse set to zero or no Mode 1 reply
**1:** X-pulse set to one (present)

Notes:

1. The flags M2, M3, MC refer to the contents of data subitems I062/120, I062/060 and I062/135 respectively. The flag M1 refers to the contents of the Subfield #5 (Extended Mode 1 Code in Octal Representation).
2. If an authenticated Mode 5 reply is received with the Emergency bit set, then the Military Emergency bit (ME) in Data Item I062/080, Track Status, shall be set.
3. If an authenticated Mode 5 reply is received with the Identification of Position bit set, then the Special Position Identification bit (SPI) in Data Item I062/080, Track Status, shall be set.
4. The resolution implied by the LSB is better than the resolution with which Mode 5 position reports are transmitted from aircraft transponders using currently defined formats.
5. GA is coded as a 14-bit two's complement binary number with an LSB of 25 ft. irrespective of the setting of RES.
6. The minimum value of GA that can be reported is -1000 ft.
7. If Subfield #1 is present, the M1 bit in Subfield #1 indicates whether the Extended Mode 1 Code is from a Mode 5 reply or a Mode 1 reply. If Subfield #1 is not present, the Extended Mode 1 Code is from a Mode 1 reply.
8. TOS shall be assumed to be zero if Subfield #6 is not present.

## I062/120 - Track Mode 2 Code

definition: Mode 2 code associated to the track

Group

Spare bits: 4

**I062/120/MODE2 - Mode-2 Code in Octal Representation**

Element
bit size: 12
Octal string (3-bits per char)

### I062/130 - Calculated Track Geometric Altitude

definition: Vertical distance between the target and the projection of its position on the earth's ellipsoid, as defined by WGS84, in two's complement form.

Element
bit size: 16
Signed quantity
LSB = $25/2^2$ ft ≈ 6.25 ft
unit: ”ft”
>= $-1500.0$
<= $150000.0$

Notes:

1. LSB is required to be less than 10 ft by ICAO
2. The source of altitude is identified in bits (SRC) of item I062/080 Track Status.

### I062/135 - Calculated Track Barometric Altitude

definition: Calculated barometric altitude of the track, in two's complement form.

Group

#### I062/135/QNH

Element
bit size: 1
Values:

**0:** No QNH correction applied
**1:** QNH correction applied

#### I062/135/CTB - Calculated Track Barometric Altitude

Element
bit size: 15
Signed quantity
LSB = $1/2^2$ FL ≈ 0.25 FL
unit: ”FL”
>= $-15.0$
<= $1500.0$

Notes:

1) ICAO specifies a range between -10 FL and 1267 FL for Mode C

### I062/136 - Measured Flight Level

definition: Last valid and credible flight level used to update the track, in two's complement form.

Element
bit size: 16
Signed quantity
LSB = $1/2^2$ FL ≈ 0.25 FL
unit: ”FL”
>= $-15.0$
<= $1500.0$

Notes:

1. The criteria to determine the credibility of the flight level are Tracker dependent.
2. Credible means: within reasonable range of change with respect to the previous detection.
3. ICAO specifies a range between -10 FL and 1267 FL for Mode C.
4. This item includes the barometric altitude received from ADS-B.

### I062/185 - Calculated Track Velocity (Cartesian)

definition: Calculated track velocity expressed in Cartesian co-ordinates,in two's complement form.

Group

#### I062/185/VX - Velocity (X-component)

Element
bit size: 16
Signed quantity
LSB = $1/2^2$ m/s ≈ $0.25$ m/s
unit: ”m/s”
>= $-8192.0$
<= $8191.75$

#### I062/185/VY - Velocity (Y-component)

Element
bit size: 16
Signed quantity
LSB = $1/2^2$ m/s ≈ $0.25$ m/s
unit: ”m/s”
>= $-8192.0$
<= $8191.75$

Notes:

- The y-axis points to the Geographical North at the location of the target.

### I062/200 - Mode of Movement

definition: Calculated Mode of Movement of a target.

Group

#### I062/200/TRANS - Transversal Acceleration

Element
bit size: 2
Values:

**0:** Constant course
**1:** Right turn
**2:** Left turn
**3:** Undetermined

#### I062/200/LONG - Longitudinal Acceleration

Element
bit size: 2
Values:

**0:** Constant groundspeed
**1:** Increasing groundspeed
**2:** Decreasing groundspeed
**3:** Undetermined

#### I062/200/VERT - Vertical Rate

Element
bit size: 2
Values:

**0:** Level
**1:** Climb
**2:** Descent
**3:** Undetermined

**I062/200/ADF - Altitude Discrepancy Flag**

Element
bit size: 1
Values:

**0:** No altitude discrepancy
**1:** Altitude discrepancy

Spare bits: 1

Notes:

- The ADF, if set, indicates that a difference has been detected in the altitude information derived from radar as compared to other technologies (such as ADS-B).

### I062/210 - Calculated Acceleration (Cartesian)

definition: Calculated Acceleration of the target expressed in Cartesian co-ordinates, in two's complement form.

Group

**I062/210/AX**

Element
bit size: 8
Signed quantity
LSB = $1/2^2$ m/s² ≈ $0.25$ m/s²
unit: ”m/s²”

**I062/210/AY**

Element
bit size: 8
Signed quantity
LSB = $1/2^2$ m/s² ≈ $0.25$ m/s²
unit: ”m/s²”

Notes:

1. The y-axis points to the Geographical North at the location of the target.
2. Maximum value means maximum value or above.

### I062/220 - Calculated Rate of Climb/Descent

definition: Calculated rate of climb/descent of an aircraft in two's complement form.

Element
bit size: 16
Signed quantity
LSB = $25/2^2$ ft/min ≈ $6.25$ ft/min
unit: ”ft/min”

Notes:

1. A positive value indicates a climb, whereas a negative value indicates a descent.

### I062/245 - Target Identification

definition: Target (aircraft or vehicle) identification in 8 characters.

Group

#### I062/245/STI

Element
bit size: 2
Values:

- **0:** Callsign or registration downlinked from target
- **1:** Callsign not downlinked from target
- **2:** Registration not downlinked from target
- **3:** Invalid

Spare bits: 6

#### I062/245/CHR - Characters 1-8 (Coded on 6 Bits Each) Defining Target Identification

Element
bit size: 48
ICAO string (6-bits per char)

Notes:

1. For coding, see section 3.1.2.9 of [Ref.3]
2. As the Callsign of the target can already be transmitted (thanks to I062/380 Subfield #2 if downlinked from the aircraft or thanks to I062/390 Subfield #2 if the target is correlated to a flight plan), and in order to avoid confusion at end user's side, this item SHALL not be used.

### I062/270 - Target Size and Orientation

definition: Target size defined as length and width of the detected target, and orientation.

Extended

#### I062/270/LENGTH - Length

Element
bit size: 7
Unsigned quantity
LSB = 1 m ≈ 1.0 m
unit: ”m”

*(FX) - extension bit*

#### I062/270/ORIENTATION - Orientation

Element
bit size: 7
Unsigned quantity
LSB = $360/2^7$ ° ≈ 2.8125 °
unit: ”°”

*(FX) - extension bit*

#### I062/270/WIDTH - Width

Element
bit size: 7
Unsigned quantity
LSB = 1 m ≈ 1.0 m
unit: ”m”

*(FX) - extension bit*

Notes:

1. The orientation gives the direction which the target nose is pointing to,relative to the Geographical North.
2. When the length only is sent, the largest dimension is provided.

### I062/290 - System Track Update Ages

definition: Ages of the last plot/local track/target report update for each sensor type.

Compound

#### I062/290/TRK - Track Age

description: Actual track age since occurence

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

#### I062/290/PSR - PSR Age

description: Age of the last primary detection used to update the track

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

#### I062/290/SSR - SSR Age

description: Age of the last secondary detection used to update the track

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

#### I062/290/MDS - Mode S Age

description: Age of the last Mode S detection used to update the track

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

#### I062/290/ADS - ADS-C Age

description: Age of the last ADS-C report used to update the track

Element
bit size: 16
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $16383.75$

#### I062/290/ES - ADS-B Extended Squitter Age

description: Age of the last 1090 Extended Squitter ADS-B report used to update the track

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/290/VDL - ADS-B VDL Mode 4 Age**

description: Age of the last VDL Mode 4 ADS-B report used to update the track

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/290/UAT - ADS-B UAT Age**

description: Age of the last UAT ADS-B report used to update the track

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/290/LOP - Loop Age**

description: Age of the last magnetic loop detection

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/290/MLT - Multilateration Age**

description: Age of the last MLT detection

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

Notes:

1. Except for Track Age, the ages are counted from Data Item I062/070, Time Of Track Information, using the following formula: Age = Time of track information - Time of last detection used to update the track
2. The time of last detection is derived from monosensor category time of day
3. If the data has never been received, then the corresponding subfield is not sent.
4. Maximum value means maximum value or above.

**I062/295 - Track Data Ages**

definition: Ages of the data provided.

Compound

**I062/295/MFL - Measured Flight Level Age**

description: Age of the last valid and credible Mode C code or barometric altitude from ADS-B used to update the track (I062/136).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/MD1 - Mode 1 Age**

description: Age of the last valid and credible Mode 1 code used to update the track (I062/110).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/MD2 - Mode 2 Age**

description: Age of the last valid and credible Mode 2 code used to update the track (I062/120).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/MDA - Mode 3/A Age**

description: Age of the last valid and credible Mode 3/A code used to update the track (I062/060).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/MD4 - Mode 4 Age**

description: Age of the last valid and credible Mode 4 code used to update the track.

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/MD5 - Mode 5 Age**

description: Age of the last valid and credible Mode 5 code used to update the track (I062/110).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/MHG - Magnetic Heading Age**

description: Age of the DAP ”Magnetic Heading” in item 062/380 (Subfield #3).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/IAS - Indicated Airspeed / Mach Nb Age**

description: Age of the DAP ”Indicated Airspeed/Mach Number” in item 062/380 (Subfield #4).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/TAS - True Airspeed Age**

description: Age of the DAP ”True Airspeed” in item 062/380 (Subfield #5).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/SAL - Selected Altitude Age**

description: Age of the DAP ”Selected Altitude” in item 062/380 (Subfield #6).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/FSS - Final State Selected Altitude Age**

description: Age of the DAP ”Final State Selected Altitude Age” in item 062/380 (Subfield #7).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/TID - Trajectory Intent Age**

description: Age of the DAP ”Trajectory Intent” in item 062/380 (Subfield #8).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/COM - Communication/ACAS Capability and Flight Status Age**

description: Age of the DAP ”Communication/ACAS Capability and Flight Status” in item 062/380 (Subfield #10).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/295/SAB - Status Reported by ADS-B Age**

description: Age of the DAP ”Status Reported by ADS-B” in item 062/380 (Subfield #11).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/295/ACS - ACAS Resolution Advisory Report Age**

description: Age of the DAP ”ACAS Resolution Advisory Report” in item 062/380 (Subfield #12).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/295/BVR - Barometric Vertical Rate Age**

description: Age of the DAP ”Barometric Vertical Rate” in item 062/380 (Subfield #13).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/295/GVR - Geometrical Vertical Rate Age**

description: Age of the DAP ”Geometrical Vertical Rate” in item 062/380 (Subfield #14).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/295/RAN - Roll Angle Age**

description: Age of the DAP ”Roll Angle” in item 062/380 (Subfield #15).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/295/TAR - Track Angle Rate Age**

description: Age of the DAP ”Track Angle Rate” in item 062/380 (Subfield #16).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/295/TAN - Track Angle Age**

description: Age of the DAP ”Track Angle” in item 062/380 (Subfield #17).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/295/GSP - Ground Speed Age**

description: Age of the DAP ”Ground Speed” in item 062/380 (Subfield #18).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/295/VUN - Velocity Uncertainty Age**

description: Age of the DAP ”Velocity Uncertainty” in item 062/380 (Subfield #19).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/295/MET - Meteorological Data Age**

description: Age of the DAP ”Meteorological Data” in item 062/380 (Subfield #20).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/295/EMC - Emitter Category Age**

description: Age of the DAP ”Emitter Category” in item 062/380 (Subfield #21).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ $0.25$ s
unit: ”s”
<= $63.75$

**I062/295/POS - Position Age**

description: Age of the DAP ”Position” in item 062/380 (Subfield #23).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/GAL - Geometric Altitude Age**

description: Age of the DAP ”Geometric Altitude” in item 062/380 (Subfield #24).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/PUN - Position Uncertainty Age**

description: Age of the DAP ”Position Uncertainty” in item 062/380 (Subfield #25).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/MB - Mode S MB Data Age**

description: Age of the DAP ”Mode S MB Data” in item 062/380 (Subfield #22).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/IAR - Indicated Airspeed Data Age**

description: Age of the DAP ”Indicated Airspeed” in item 062/380 (Subfield #26).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/MAC - Mach Number Data Age**

description: Age of the DAP ”Mach Number” in item 062/380 (Subfield #27).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: ”s”
<= 63.75

**I062/295/BPS - Barometric Pressure Setting Data Age**

description: Age of the DAP ”Barometric Pressure Setting” in item 062/380 (Subfield #28).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ s ≈ 0.25 s
unit: "s"
<= 63.75

Notes:

1. Despite there are now two subfields (#29 and #30) reporting the ages of, respectively, the Indicated Airspeed track data and the Mach Number track data, the subfield #8 (and so its presence bit , bit-32) is kept free in order to prevent a full incompatibility with previous releases of ASTERIX Cat. 062 already implemented.
2. In all the subfields, the age is the time delay since the value was measured

### I062/300 - Vehicle Fleet Identification

definition: Vehicle fleet identification number.

Element
bit size: 8
Values:

**0:** Unknown
**1:** ATC equipment maintenance
**2:** Airport maintenance
**3:** Fire
**4:** Bird scarer
**5:** Snow plough
**6:** Runway sweeper
**7:** Emergency
**8:** Police
**9:** Bus
**10:** Tug (push/tow)
**11:** Grass cutter
**12:** Fuel
**13:** Baggage
**14:** Catering
**15:** Aircraft maintenance
**16:** Flyco (follow me)

### I062/340 - Measured Information

definition: All measured data related to the last report used to update the track. These data are not used for ADS-B.

Compound

#### I062/340/SID - Sensor Identification

Group

##### I062/340/SID/SAC - System Area Code

Element
bit size: 8
Raw Content

##### I062/340/SID/SIC - System Identification Code

Element
bit size: 8
Raw Content

#### I062/340/POS - Measured Position

Group

**I062/340/POS/RHO - Measured Distance**

Element
bit size: 16
Unsigned quantity
LSB = $1/2^{8}$ NM ≈ $3.90625e-3$ NM
unit: ”NM”
<= 256.0

**I062/340/POS/THETA - Measured Azimuth**

Element
bit size: 16
Unsigned quantity
LSB = $360/2^{1}6$ ° ≈ $5.4931640625e-3$ °
unit: ”°”

**I062/340/HEIGHT - Measured 3-D Height**

Element
bit size: 16
Signed quantity
LSB = 25 ft ≈ 25.0 ft
unit: ”ft”

The reference level for this height information is the same as the reference level applied by the sensor system providing this information.

**I062/340/MDC**

Group

**I062/340/MDC/V - Validated**

Element
bit size: 1
Values:

**0:** Code validated
**1:** Code not validated

**I062/340/MDC/G - Garbled**

Element
bit size: 1
Values:

**0:** Default
**1:** Garbled code

**I062/340/MDC/LMC - Last Measured Mode C Code**

description: Last Measured Mode C Code, in two's complement form

Element
bit size: 14
Signed quantity
LSB = $1/2^{2}$ FL ≈ 0.25 FL
unit: ”FL”
>= −12.0
<= 1270.0

**I062/340/MDA**

Group

**I062/340/MDA/V - Validated**

Element
bit size: 1
Values:

**0:** Code validated
**1:** Code not validated

**I062/340/MDA/G - Garbled**

Element
bit size: 1
Values:

**0:** Default
**1:** Garbled code

**I062/340/MDA/L**

Element
bit size: 1
Values:

**0:** Mode 3/A code as derived from the reply of the transponder
**1:** Mode 3/A code as provided by a sensor local tracker

Spare bits: 1

**I062/340/MDA/MODE3A - Mode-3/A Reply in Octal Representation**

Element
bit size: 12
Octal string (3-bits per char)

**I062/340/TYP**

Group

**I062/340/TYP/TYP - Report Type**

Element
bit size: 3
Values:

**0:** No detection
**1:** Single PSR detection
**2:** Single SSR detection
**3:** SSR + PSR detection
**4:** Single ModeS All-Call
**5:** Single ModeS Roll-Call
**6:** ModeS All-Call + PSR
**7:** ModeS Roll-Call + PSR

**I062/340/TYP/SIM**

Element
bit size: 1
Values:

**0:** Actual target report
**1:** Simulated target report

**I062/340/TYP/RAB**

Element
bit size: 1
Values:

**0:** Report from target transponder
**1:** Report from field monitor (item transponder)

**I062/340/TYP/TST**

Element
bit size: 1
Values:

**0:** Real target report
**1:** Test target report

Spare bits: 2

Notes:

1. In case of a plot, the measured bias-corrected polar co-ordinates;
2. In case of a sensor local track, the measured bias-corrected polar co-ordinates of the plot associated to the track;
3. In case of a local track without detection, the extrapolated bias-corrected polar co-ordinates.
4. Smoothed MODE 3/A data (L = 1) will be used in case of absence of MODE 3/A code information in the plot or in case of difference between plot and sensor local track MODE 3/A code information.

### I062/380 - Aircraft Derived Data

definition: Data derived directly by the aircraft.

Compound

#### I062/380/ADR - Target Address

Element
bit size: 24
Raw Content

#### I062/380/ID - Target Identification

description: Characters 1-8 (coded on 6 bits each) defining a target identification when flight plan is available or the registration marking when no flight plan is available. Coding rules are provided in [3] Section 3.1.2.9.1.2 and Table 3-9

Element
bit size: 48
ICAO string (6-bits per char)

#### I062/380/MHG - Magnetic Heading

Element
bit size: 16
Unsigned quantity
LSB = $360/2^16$ ° ≈ $5.4931640625e-3$ °
unit: "°"

#### I062/380/IAS - Indicated Airspeed/Mach No

Group

##### I062/380/IAS/IM

Element
bit size: 1
Values:

**0:** Air Speed = IAS, LSB (Bit-1) = 2^-14 NM/s
**1:** Air Speed = Mach, LSB (Bit-1) = 0.001

##### I062/380/IAS/IAS

Element
bit size: 15
Depending on: (`380/IAS/IM`)

**(0):** Unsigned quantity
LSB = $1/2^14$ NM/s ≈ $6.103515625e-5$ NM/s
unit: "NM/s"

**(1):** Unsigned quantity
LSB = $1/1000$ Mach ≈ $1.0e-3$ Mach
unit: "Mach"

Default:

Raw Content

#### I062/380/TAS - True Airspeed

Element
bit size: 16
Unsigned quantity
LSB = 1 kt ≈ 1.0 kt
unit: ”kt”
>= 0.0
<= 2046.0

**I062/380/SAL - Selected Altitude**

Group

**I062/380/SAL/SAS**

Element
bit size: 1
Values:

**0:** No source information provided
**1:** Source information provided

**I062/380/SAL/SRC**

Element
bit size: 2
Values:

**0:** Unknown
**1:** Aircraft altitude
**2:** FCU/MCP selected altitude
**3:** FMS selected altitude

**I062/380/SAL/ALT - Altitude in Two's Complement Form**

Element
bit size: 13
Signed quantity
LSB = 25 ft ≈ 25.0 ft
unit: ”ft”
>= −1300.0
<= 100000.0

**I062/380/FSS - Final State Selected Altitude**

Group

**I062/380/FSS/MV - Manage Vertical Mode**

description: Manage Vertical Mode

Element
bit size: 1
Values:

**0:** Not active
**1:** Active

**I062/380/FSS/AH - Altitude Hold**

description: Altitude Hold

Element
bit size: 1
Values:

**0:** Not active
**1:** Active

**I062/380/FSS/AM - Approach Mode**

description: Approach Mode

Element
bit size: 1
Values:

**0:** Not active
**1:** Active

**I062/380/FSS/ALT - Altitude in Two's Complement Form**

Element
bit size: 13
Signed quantity
LSB = 25 ft ≈ 25.0 ft
unit: ”ft”
>= −1300.0
<= 100000.0

**I062/380/TIS - Trajectory Intent Status**

Extended

**I062/380/TIS/NAV - TID Available**

Element
bit size: 1
Values:

**0:** Trajectory intent data is available for this aircraft
**1:** Trajectory intent data is not available for this aircraft

**I062/380/TIS/NVB - TID Valid**

Element
bit size: 1
Values:

**0:** Trajectory intent data is valid
**1:** Trajectory intent data is not valid

Spare bits: 5
*(FX) - extension bit*

**I062/380/TID - Trajectory Intent Data**

Repetitive
Regular, 1 byte(s) REP field size.
Group

**I062/380/TID/TCA - TCP Number Availability**

Element
bit size: 1
Values:

**0:** TCP number available
**1:** TCP number not available

**I062/380/TID/NC - TCP Compliance**

Element
bit size: 1
Values:

**0:** TCP compliance
**1:** TCP non-compliance

**I062/380/TID/TCPN - Trajectory Change Point Number**

description: Trajectory change point number
Element
bit size: 6
Raw Content

**I062/380/TID/ALT - Altitude in Two's Complement Form**

Element
bit size: 16
Signed quantity
LSB = 10 ft ≈ 10.0 ft
unit: ”ft”
>= −1500.0
<= 150000.0

**I062/380/TID/LAT - Latitude in WGS.84 in Two's Complement**

Element
bit size: 24
Signed quantity
LSB = $180/2^{2}3$ ° ≈ $2.1457672119140625e-5$ °
unit: ”°”
>= $-90.0$
<= $90.0$

**I062/380/TID/LON - Longitude in WGS.84 in Two's Complement**

Element
bit size: 24
Signed quantity
LSB = $180/2^{2}3$ ° ≈ $2.1457672119140625e-5$ °
unit: ”°”
>= $-180.0$
< $180.0$

**I062/380/TID/PT - Point Type**

Element
bit size: 4
Values:

**0:** Unknown
**1:** Fly by waypoint (LT)
**2:** Fly over waypoint (LT)
**3:** Hold pattern (LT)
**4:** Procedure hold (LT)
**5:** Procedure turn (LT)
**6:** RF leg (LT)
**7:** Top of climb (VT)
**8:** Top of descent (VT)
**9:** Start of level (VT)
**10:** Cross-over altitude (VT)
**11:** Transition altitude (VT)

**I062/380/TID/TD - Turn Direction**

Element
bit size: 2
Values:

**0:** N/A
**1:** Turn right
**2:** Turn left
**3:** No turn

**I062/380/TID/TRA - Turn Radius Availability**

description: Turn Radius Availability
Element
bit size: 1
Values:

**0:** TTR not available
**1:** TTR available

**I062/380/TID/TOA - TOV Available**

Element
bit size: 1
Values:

**0:** TOV available
**1:** TOV not available

**I062/380/TID/TOV - Time Over Point**

Element
bit size: 24
Unsigned quantity

LSB = 1 s ≈ 1.0 s
unit: ”s”

**I062/380/TID/TTR - TCP Turn Radius**

Element
bit size: 16
Unsigned quantity
LSB = $1/100$ NM ≈ $1.0e-2$ NM
unit: ”NM”
>= 0.0
<= 655.35

**I062/380/COM - Communications/ACAS Capability and Flight Status**

Group

**I062/380/COM/COM - Communications Capability of the Transponder**

Element
bit size: 3
Values:

- **0:** No communications capability (surveillance only)
- **1:** Comm. A and Comm. B capability
- **2:** Comm. A, Comm. B and Uplink ELM
- **3:** Comm. A, Comm. B, Uplink ELM and Downlink ELM
- **4:** Level 5 Transponder capability
- **5:** Not assigned
- **6:** Not assigned
- **7:** Not assigned

**I062/380/COM/STAT - Flight Status**

Element
bit size: 3
Values:

- **0:** No alert, no SPI, aircraft airborne
- **1:** No alert, no SPI, aircraft on ground
- **2:** Alert, no SPI, aircraft airborne
- **3:** Alert, no SPI, aircraft on ground
- **4:** Alert, SPI, aircraft airborne or on ground
- **5:** No alert, SPI, aircraft airborne or on ground
- **6:** Not defined
- **7:** Unknown or not yet extracted

Spare bits: 2

**I062/380/COM/SSC - Specific Service Capability**

Element
bit size: 1
Values:

- **0:** No
- **1:** Yes

**I062/380/COM/ARC - Altitude Reporting Capability**

Element
bit size: 1
Values:

- **0:** 100 ft resolution
- **1:** 25 ft resolution

**I062/380/COM/AIC - Aircraft Identification Capability**

Element
bit size: 1
Values:

**0:** No
**1:** Yes

**I062/380/COM/B1A - BDS 1,0 Bit 16**

Element
bit size: 1
Raw Content

**I062/380/COM/B1B - BDS BDS 1,0 Bits 37/40**

Element
bit size: 4
Raw Content

**I062/380/SAB - Status Reported by ADS-B**

Group

**I062/380/SAB/AC - ACAS Status**

Element
bit size: 2
Values:

**0:** Unknown
**1:** ACAS not operational
**2:** ACAS operational
**3:** Invalid

**I062/380/SAB/MN - Multiple Navigational Aids Status**

Element
bit size: 2
Values:

**0:** Unknown
**1:** Multiple navigational aids not operating
**2:** Multiple navigational aids operating
**3:** Invalid

**I062/380/SAB/DC - Differential Correction Status**

Element
bit size: 2
Values:

**0:** Unknown
**1:** Differential correction
**2:** No differential correction
**3:** Invalid

**I062/380/SAB/GBS - Ground Bit Set**

Element
bit size: 1
Values:

**0:** Transponder ground bit not set or unknown
**1:** Transponder Ground Bit set

Spare bits: 6

**I062/380/SAB/STAT - Flight Status**

Element
bit size: 3
Values:

**0:** No emergency
**1:** General emergency
**2:** Lifeguard / medical
**3:** Minimum fuel
**4:** No communications
**5:** Unlawful interference
**6:** Downed Aircraft
**7:** Unknown

**I062/380/ACS - ACAS Resolution Advisory Report**

description: Currently active Resolution Advisory (RA), if any, generated by the ACAS associated with the transponder transmitting the report and threat identity data. (MB Data) 56-bit message conveying Mode S Comm B message data of BDS Register 3,0

Element
bit size: 56
BDS register at address 48

**I062/380/BVR - Barometric Vertical Rate**

description: Barometric Vertical Rate in two's complement form

Element
bit size: 16
Signed quantity
LSB = $25/2^2$ ft/min ≈ $6.25$ ft/min
unit: ”ft/min”

**I062/380/GVR - Geometric Vertical Rate**

description: Geometric Vertical Rate in two's complement form

Element
bit size: 16
Signed quantity
LSB = $25/2^2$ ft/min ≈ $6.25$ ft/min
unit: ”ft/min”

**I062/380/RAN - Roll Angle**

description: Roll Angle in two's complement form

Element
bit size: 16
Signed quantity
LSB = $1/100$ ° ≈ $1.0e-2$ °
unit: ”°”
>= $-180.0$
<= $180.0$

**I062/380/TAR - Track Angle Rate**

Group

**I062/380/TAR/TI - Turn Indicator**

Element
bit size: 2
Values:

**0:** Not available
**1:** Left
**2:** Right
**3:** Straight

Spare bits: 6

**I062/380/TAR/ROT - Rate of Turn in Two's Complement Form**

Element
bit size: 7
Signed quantity
LSB = $1/2^2$ °/s ≈ $0.25$ °/s
unit: ”°/s”
>= $-15.0$
<= $15.0$

Spare bits: 1

**I062/380/TAN - Track Angle**

Element
bit size: 16
Unsigned quantity
LSB = $360/2^16$ ° ≈ $5.4931640625e-3$ °
unit: ”°”

**I062/380/GS - Ground Speed**

description: Ground Speed in Two's Complement Form Referenced to WGS84

Element
bit size: 16
Signed quantity
LSB = $1/2^14$ NM/s ≈ $6.103515625e-5$ NM/s
unit: ”NM/s”
>= $-2.0$
< $2.0$

**I062/380/VUN - Velocity Uncertainty**

Element
bit size: 8
Raw Content

**I062/380/MET - Meteorological Data**

Group

**I062/380/MET/WS - Wind Speed Valid Flag**

Element
bit size: 1
Values:
**0:** Not valid Wind Speed
**1:** Valid Wind Speed

**I062/380/MET/WD - Wind Direction Valid Flag**

Element
bit size: 1
Values:
**0:** Not valid Wind Direction
**1:** Valid Wind Direction

**I062/380/MET/TMP - Temperature Valid Flag**

Element
bit size: 1
Values:
**0:** Not valid Temperature
**1:** Valid Temperature

**I062/380/MET/TRB - Turbulence Valid Flag**

Element
bit size: 1
Values:
**0:** Not valid Turbulence
**1:** Valid Turbulence

Spare bits: 4

**I062/380/MET/WSD - Wind Speed**

Element
bit size: 16
Unsigned quantity
LSB = 1 kt ≈ 1.0 kt
unit: ”kt”
>= 0.0
<= 300.0

**I062/380/MET/WDD - Wind Direction**

Element
bit size: 16
Unsigned quantity
LSB = 1 ° ≈ 1.0 °
unit: ”°”
>= 1.0
<= 360.0

#### I062/380/MET/TMPD - Temperature in Degrees Celsius

Element
bit size: 16
Signed quantity
LSB = $1/2^2$ °C ≈ 0.25 °C
unit: ”°C”
>= $-100.0$
<= $100.0$

#### I062/380/MET/TRBD - Turbulence

Element
bit size: 8
Unsigned integer
>= $0.0$
<= $15.0$

### I062/380/EMC - Emitter Category

Element
bit size: 8
Values:

- **1:** Light aircraft =< 7000 kg
- **2:** Reserved
- **3:** 7000 kg < medium aircraft < 136000 kg
- **4:** Reserved
- **5:** 136000 kg <= heavy aircraft
- **6:** Highly manoeuvrable (5g acceleration capability) and high speed (>400 knots cruise)
- **7:** Reserved
- **8:** Reserved
- **9:** Reserved
- **10:** Rotocraft
- **11:** Glider / sailplane
- **12:** Lighter-than-air
- **13:** Unmanned aerial vehicle
- **14:** Space / transatmospheric vehicle
- **15:** Ultralight / handglider / paraglider
- **16:** Parachutist / skydiver
- **17:** Reserved
- **18:** Reserved
- **19:** Reserved
- **20:** Surface emergency vehicle
- **21:** Surface service vehicle
- **22:** Fixed ground or tethered obstruction
- **23:** Reserved
- **24:** Reserved

### I062/380/POS - Position

Group

#### I062/380/POS/LAT - Latitude in WGS.84 in Two's Complement Form

Element
bit size: 24
Signed quantity
LSB = $180/2^23$ ° ≈ $2.1457672119140625e-5$ °
unit: ”°”
>= $-90.0$
<= $90.0$

#### I062/380/POS/LON - Longitude in WGS.84 in Two's Complement Form

Element

bit size: 24
Signed quantity
LSB = $180/2^23$ ° ≈ $2.1457672119140625e-5$ °
unit: ”°”
>= $-180.0$
< $180.0$

This corresponds to a resolution of at least 2.4 meters.

**I062/380/GAL - Geometric Altitude**

Element
bit size: 16
Signed quantity
LSB = $25/2^2$ ft ≈ $6.25$ ft
unit: ”ft”
>= $-1500.0$
<= $150000.0$

**I062/380/PUN - Position Uncertainty**

Group

Spare bits: 4

**I062/380/PUN/PUN - Position Uncertainty**

Element
bit size: 4
Raw Content

**I062/380/BDSDATA - BDS Register DATA**

Repetitive
Regular, 1 byte(s) REP field size.

Element
bit size: 64
BDS register with address

**I062/380/IAR - Indicated Airspeed**

Element
bit size: 16
Unsigned quantity
LSB = 1 kt ≈ $1.0$ kt
unit: ”kt”
>= $0.0$
<= $1100.0$

**I062/380/MAC - Mach Number**

Element
bit size: 16
Unsigned quantity
LSB = $1/125$ Mach ≈ $8.0e-3$ Mach
unit: ”Mach”
>= $0.0$
<= $4.096$

**I062/380/BPS - Barometric Pressure Setting**

Group

Spare bits: 4

**I062/380/BPS/BPS**

Element
bit size: 12
Unsigned quantity
LSB = $1/10$ mb ≈ $0.1$ mb
unit: ”mb”
>= $0.0$
<= $409.5$

Notes:

1. NC is set to one when the aircraft will not fly the path described by the TCP data.
2. TCP numbers start from zero.
3. LT = Lateral Type
4. VT = Vertical Type
5. TOV gives the estimated time before reaching the point. It is defined as the absolute time from midnight.
6. TOV is meaningful only if TOA is set to 0
7. To bits 3/1 (STAT): For ADS-B Version 3 systems as defined in ED-102B/DO-260C (Ref. [11]) the values have been re-defined.
8. I062/REF/PS3 is to be used exclusively for Version 3 ADS-B systems as defined in I062/380/SF#11/VN. For ADS-B systems with a version number below 3, the PS shall be encoded in Data Item I062/380 SF#11/STAT.
9. In case of an ADS-B Version 3 system as defined in ED-102B/DO-260C (Ref. [11]) in order to maintain backwards compatibility also I062/380/SF#11/STAT shall be populated. However, since values have been re-defined in ADS-B Version 3, mapping is required to ensure that information is not lost. This mapping shall be done according to the following table: :

| ADS-B Version 3 (I062/REF/PS3) | ADS-Version < 3 (I062/380/SF#11/STAT) |
|---|---|
| 0 (No Emergency/not reported) | 0 (No Emergency/not reported) |
| 1 (General emergency) | 1 (General emergency) |
| 2 (UAS/RPAS Lost Link) | 4 (No communication) |
| 3 (Minimum fuel) | 3 (Minimum fuel) |
| 4 (No communication) | 4 (No communication) |
| 5 (Unlawful interference) | 5 (Unlawful interference) |
| 6 (Aircraft in distress - automatic activation) | 1 (General emergency) |
| 7 (Aircraft in distress - manual activation) | 1 (General emergency) |

10. Refer to ICAO Draft SARPs for ACAS for detailed explanations.
11. A positive value represents a right turn, whereas a negative value represents a left turn.
12. Value 15 means 15 degrees/s or above.
13. Velocity uncertainty category of the least accurate velocity component
14. Positive longitude indicates East. Positive latitude indicates North.
15. LSB is required to be thinner than 10 ft by ICAO
16. Only DAPs that can not be encoded into other subfields of this item should be sent using subfield #25
17. BPS is the barometric pressure setting of the aircraft minus 800 mb.
18. As of Edition 1.19 the note ”(derived from Mode S BDS 4,0)” has been removed to allow transmission of BPS received via ADS-B.

### I062/390 - Flight Plan Related Data

definition: All flight plan related information, provided by ground-based systems.

Compound

#### I062/390/TAG - FPPS Identification Tag

Group

##### I062/390/TAG/SAC - System Area Code

Element
bit size: 8
Raw Content

**I062/390/TAG/SIC - System Identification Code**

Element
bit size: 8
Raw Content

**I062/390/CS - Callsign**

Element
bit size: 56
Ascii string (8-bits per char)

**I062/390/IFI - IFPS_FLIGHT_ID**

Group

**I062/390/IFI/TYP**

Element
bit size: 2
Values:

- **0:** Plan Number
- **1:** Unit 1 internal flight number
- **2:** Unit 2 internal flight number
- **3:** Unit 3 internal flight number

Spare bits: 3

**I062/390/IFI/NBR - Number from 0 to 99 999 999**

Element
bit size: 27
Unsigned integer
>= $0.0$
<= $9.9999999e7$

**I062/390/FCT - Flight Category**

Group

**I062/390/FCT/GATOAT**

Element
bit size: 2
Values:

- **0:** Unknown
- **1:** General Air Traffic
- **2:** Operational Air Traffic
- **3:** Not applicable

**I062/390/FCT/FR1FR2**

Element
bit size: 2
Values:

- **0:** Instrument Flight Rules
- **1:** Visual Flight Rules
- **2:** Not applicable
- **3:** Controlled Visual Flight Rules

**I062/390/FCT/RVSM**

Element
bit size: 2
Values:

- **0:** Unknown
- **1:** Approved
- **2:** Exempt
- **3:** Not Approved

**I062/390/FCT/HPR**

Element
bit size: 1
Values:

**0:** Normal Priority Flight
**1:** High Priority Flight

Spare bits: 1

**I062/390/TAC - Type of Aircraft**

Element
bit size: 32
Ascii string (8-bits per char)

**I062/390/WTC - Wake Turbulence Category**

Element
bit size: 8
Ascii string (8-bits per char)

**I062/390/DEP - Departure Airport**

Element
bit size: 32
Ascii string (8-bits per char)

**I062/390/DST - Destination Airport**

Element
bit size: 32
Ascii string (8-bits per char)

**I062/390/RDS - Runway Designation**

Group

**I062/390/RDS/NU1 - First Number**

Element
bit size: 8
Ascii string (8-bits per char)

**I062/390/RDS/NU2 - Second Number**

Element
bit size: 8
Ascii string (8-bits per char)

**I062/390/RDS/LTR - Letter**

Element
bit size: 8
Ascii string (8-bits per char)

**I062/390/CFL - Current Cleared Flight Level**

Element
bit size: 16
Unsigned quantity
LSB = $1/2^2$ FL ≈ 0.25 FL
unit: ”FL”
< 1500.0

**I062/390/CTL - Current Control Position**

Group

**I062/390/CTL/CENTRE - 8-bit Group Identification Code**

Element
bit size: 8
Raw Content

**I062/390/CTL/POSITION - 8-bit Control Position Identification Code**

Element
bit size: 8
Raw Content

**I062/390/TOD - Time of Departure / Arrival**

Repetitive
Regular, 1 byte(s) REP field size.
Group

**I062/390/TOD/TYP**

Element
bit size: 5
Values:

- **0:** Scheduled off-block time
- **1:** Estimated off-block time
- **2:** Estimated take-off time
- **3:** Actual off-block time
- **4:** Predicted time at runway hold
- **5:** Actual time at runway hold
- **6:** Actual line-up time
- **7:** Actual take-off time
- **8:** Estimated time of arrival
- **9:** Predicted landing time
- **10:** Actual landing time
- **11:** Actual time off runway
- **12:** Predicted time to gate
- **13:** Actual on-block time

**I062/390/TOD/DAY**

Element
bit size: 2
Values:

- **0:** Today
- **1:** Yesterday
- **2:** Tomorrow
- **3:** Invalid

Spare bits: 4

**I062/390/TOD/HOR - Hours**

Element
bit size: 5
Unsigned integer
>= 0.0
<= 23.0

Spare bits: 2

**I062/390/TOD/MIN - Minutes**

Element
bit size: 6
Unsigned integer
>= 0.0
<= 59.0

**I062/390/TOD/AVS - Seconds Available Flag**

Element
bit size: 1
Values:

- **0:** Seconds available
- **1:** Seconds not available

Spare bits: 1

**I062/390/TOD/SEC - Seconds**

Element
bit size: 6
Unsigned integer
>= 0.0
<= 59.0

**I062/390/AST - Aircraft Stand**

Element
bit size: 48
Ascii string (8-bits per char)

**I062/390/STS - Stand Status**

Group

**I062/390/STS/EMP**

Element
bit size: 2
Values:

- **0:** Empty
- **1:** Occupied
- **2:** Unknown
- **3:** Invalid

**I062/390/STS/AVL**

Element
bit size: 2
Values:

- **0:** Available
- **1:** Not available
- **2:** Unknown
- **3:** Invalid

Spare bits: 4

**I062/390/STD - Standard Instrument Departure**

Element
bit size: 56
Ascii string (8-bits per char)

**I062/390/STA - Standard Instrument Arrival**

Element
bit size: 56
Ascii string (8-bits per char)

**I062/390/PEM - Pre-Emergency Mode 3/A**

Group

Spare bits: 3

**I062/390/PEM/VA**

Element
bit size: 1
Values:

- **0:** No valid Mode 3/A available
- **1:** Valid Mode 3/A available

**I062/390/PEM/MODE3A - Mode-3/A Reply in Octal Representation**

Element
bit size: 12
Octal string (3-bits per char)

**I062/390/PEC - Pre-Emergency Callsign**

Element
bit size: 56
Ascii string (8-bits per char)

Notes:

1. The up-to-date list of SACs is published on the Eurocontrol Web Site (http://www.eurocontrol.int).

2. Each one of the seven Octets contains an ASCII Character. TheCallsign is always left adjusted. It contains up to seven upper-case alphanumeric characters, the remaining character positions (if any)are padded with space characters.
3. Each one of the four Octets composing the type of an aircraft contains an ASCII Character (upper-case alphanumeric characters with trailing spaces).
4. The types of aircraft are defined in [Ref.4]
5. Each one of the four Octets composing the name of an airport contains an ASCII Character (upper case alphabetic).
6. The Airport Names are indicated in the ICAO Location Indicators book.
7. Each one of the four Octets composing the name of an airport contains an ASCII Character (upper case alphabetic).
8. The Airport Names are indicated in the ICAO Location Indicators book.
9. NU1, NU2 and LTR each contain an ASCII character
10. For details refer to.[5] Section 5
11. The centre and the control position identification codes have to be defined between communication partners.
12. Estimated times are derived from flight plan systems. Predicted times are derived by the fusion system, based on surveillance data. For definitions, see [Ref.4]
13. Each one of the six Octets contains an ASCII Character. The Aircraft Stand identification is always left adjusted. It contains up to six upper-case alphanumeric characters, the remaining character positions (if any) are padded with space characters.
14. Each one of the seven Octets contains an ASCII Character. The SID is always left adjusted. It contains up to seven alphanumeric characters, the remaining character positions (if any) are padded with space characters.
15. Each one of the seven Octets contains an ASCII Character. The STAR is always left adjusted. It contains up to seven alphanumeric characters, the remaining character positions (if any) are padded with space characters.
16. This subfield is used only when the aircraft is transmitting an emergency Mode 3/A code
17. If VA = 0, the content of bits 12/1 is meaningless
18. Each one of the seven Octets contains an ASCII Character. The Callsign is always left adjusted. It contains up to seven upper-case alphanumeric characters, the remaining character positions (if any) are padded with space characters
19. This subfield is used only when an emergency Mode 3/A is associated with the track (I062/390 Subfield #17)

**I062/500 - Estimated Accuracies**

definition: Overview of all important accuracies.

Compound

**I062/500/APC - Estimated Accuracy Of Track Position (Cartesian)**

Group

**I062/500/APC/X - APC (X-Component)**

Element
bit size: 16
Unsigned quantity
LSB = $1/2$ m $\approx 0.5$ m
unit: ”m”

**I062/500/APC/Y - APC (Y-Component)**

Element
bit size: 16
Unsigned quantity
LSB = $1/2$ m $\approx 0.5$ m
unit: ”m”

**I062/500/COV - XY Covariance Component**

Element
bit size: 16
Signed quantity
LSB = $1/2$ m ≈ $0.5$ m
unit: ”m”

**I062/500/APW - Estimated Accuracy Of Track Position (WGS-84)**

Group

**I062/500/APW/LAT - APW (Latitude Component)**

Element
bit size: 16
Unsigned quantity
LSB = $180/2^25$ ° ≈ $5.36441802978515625e-6$ °
unit: ”°”

**I062/500/APW/LON - APW (Longitude Component)**

Element
bit size: 16
Unsigned quantity
LSB = $180/2^25$ ° ≈ $5.36441802978515625e-6$ °
unit: ”°”

**I062/500/AGA - Estimated Accuracy Of Calculated Track Geometric Altitude**

Element
bit size: 8
Unsigned quantity
LSB = $25/2^2$ ft ≈ $6.25$ ft
unit: ”ft”

**I062/500/ABA - Estimated Accuracy Of Calculated Track Barometric Altitude**

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ FL ≈ $0.25$ FL
unit: ”FL”

**I062/500/ATV - Estimated Accuracy Of Track Velocity (Cartesian)**

Group

**I062/500/ATV/X - ATV (X-Component)**

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ m/s ≈ $0.25$ m/s
unit: ”m/s”

**I062/500/ATV/Y - ATV (Y-Component)**

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ m/s ≈ $0.25$ m/s
unit: ”m/s”

**I062/500/AA - Estimated Accuracy Of Acceleration (Cartesian)**

Group

**I062/500/AA/X - AA (X-Component)**

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ m/s² ≈ $0.25$ m/s²
unit: ”m/s²”

**I062/500/AA/Y - AA (Y-Component)**

Element
bit size: 8
Unsigned quantity
LSB = $1/2^2$ m/s² ≈ $0.25$ m/s²
unit: ”m/s²”

**I062/500/ARC - Estimated Accuracy Of Rate Of Climb/Descent**

Element
bit size: 8
Unsigned quantity
LSB = $25/2^2$ ft/min ≈ $6.25$ ft/min
unit: ”ft/min”

Notes:

1. Maximum value means maximum value or above.
2. XY covariance component = sign {Cov(X,Y)} * sqrt {abs [Cov (X,Y)]}
3. The maximum value for the (unsigned) XY covariance component is 16.383 km
4. Maximum value means maximum value or above.
5. Maximum value means maximum value or above.
6. Maximum value means maximum value or above.
7. Maximum value means maximum value or above.
8. Maximum value means maximum value or above.
9. Maximum value means maximum value or above.

**I062/510 - Composed Track Number**

definition: Identification of a system track.

Repetitive

With FX extension bit.

Group

**I062/510/IDENT - System Unit Identification**

Element
bit size: 8
Raw Content

**I062/510/TRACK - System Track Number**

Element
bit size: 15
Raw Content

Notes:

- The composed track number is used by co-operating units to uniquely identify a track. It consists of the unit identifier and system track number for each unit involved in the co-operation. The first unit identification identifies the unit that is responsible for the track amalgamation.
- The fist element represents Master track, the remaining elements represent Slave tracks.

**I062/RE - Reserved Expansion Field**

definition: Expansion

Explicit (ReservedExpansion)

**I062/SP - Special Purpose Field**

definition: Special Purpose Field

Explicit (SpecialPurpose)

## User Application Profile

- 1: `I062/010` - Data Source Identifier
- *Spare*
- 3: `I062/015` - Service Identification
- 4: `I062/070` - Time Of Track Information
- 5: `I062/105` - Calculated Position In WGS-84 Co-ordinates
- 6: `I062/100` - Calculated Track Position (Cartesian)
- 7: `I062/185` - Calculated Track Velocity (Cartesian)
- (FX) - Field extension indicator
- 8: `I062/210` - Calculated Acceleration (Cartesian)
- 9: `I062/060` - Track Mode 3/A Code
- 10: `I062/245` - Target Identification
- 11: `I062/380` - Aircraft Derived Data
- 12: `I062/040` - Track Number
- 13: `I062/080` - Track Status
- 14: `I062/290` - System Track Update Ages
- (FX) - Field extension indicator
- 15: `I062/200` - Mode of Movement
- 16: `I062/295` - Track Data Ages
- 17: `I062/136` - Measured Flight Level
- 18: `I062/130` - Calculated Track Geometric Altitude
- 19: `I062/135` - Calculated Track Barometric Altitude
- 20: `I062/220` - Calculated Rate of Climb/Descent
- 21: `I062/390` - Flight Plan Related Data
- (FX) - Field extension indicator
- 22: `I062/270` - Target Size and Orientation
- 23: `I062/300` - Vehicle Fleet Identification
- 24: `I062/110` - Mode 5 Data Reports and Extended Mode 1 Code
- 25: `I062/120` - Track Mode 2 Code
- 26: `I062/510` - Composed Track Number
- 27: `I062/500` - Estimated Accuracies
- 28: `I062/340` - Measured Information
- (FX) - Field extension indicator
- *Spare*
- *Spare*
- *Spare*
- *Spare*
- *Spare*
- 34: `I062/RE` - Reserved Expansion Field
- 35: `I062/SP` - Special Purpose Field
- (FX) - Field extension indicator